

Total No. of Questions : 8]

[Total No. of Printed Pages : 1

Roll No

EC-804 (GS)
B.E. VIII Semester Examination, June 2020
Grading System (GS)
TV and Radar Engineering
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. How many frames are scanned per second? Support your answer with proper justification.
2. a) With the help of block diagram explain the basic principle of sound and picture transmission.
b) Explain flicker. How it can be removed?
3. Explain the following :
 - i) Aspect ratio
 - ii) Interlaced scanning
 - iii) Picture & sound modulation
4. What are the essential differences between an NTSC system and a PAL system? Why the aspect ratio is maintained at 4:3?
5. Define and explain the terms
 - i) Primary colours
 - ii) Complementary colours
 - iii) Hue
 - iv) Saturation
6. Write short notes on:
 - a) CCTV
 - b) LCD display
 - c) Synthetic aperture radar
7. Calculate the maximum range of Radar in which operate at 3cm with a peak pulse power of 500 kW, its minimum receivable power is 10^{-13} W, the capture area of its antenna is 5m^2 and target cross section area is of 20m^2 .
8. Write short notes on any two of the following:
 - i) Mixing of colours and chromaticity diagram.
 - ii) Colour Television camera
 - iii) Colour TV receiver
 - iv) High Definition Television system (HDTV)
 - v) MTI Radar
